AgriIntel

Yield, profit and supply-aware crop advisory

Data Science Programming
Course Project Final Review

Manish H 24MID0294
Anshul Singh 24MID0291
Aryan Sinha 24MID0244

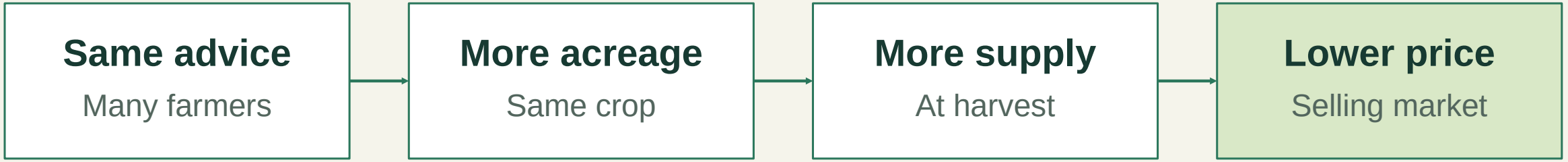
Batch 22
Faculty: Padma Priya M

Regional crop advisory prototype

Crop choice and market oversupply

Intended users: farmers and regional agricultural advisers

A crop can look profitable before planting, then lose value when many farmers choose it.



Yield is output per hectare. Oversupply primarily affects the selling price.

Project question: can crop advice account for both uncertainty and planting commitments?

Objectives and scope

Five-state historical data with a bounded regional feedback demonstration

01 **Forecast yield uncertainty**

Report lower, median and upper yield estimates.

02 **Explain financial scenarios**

Show costs, price assumptions and downside scenarios.

03 **Coordinate voluntary commitments**

Accept planting area only within defined capacity constraints.

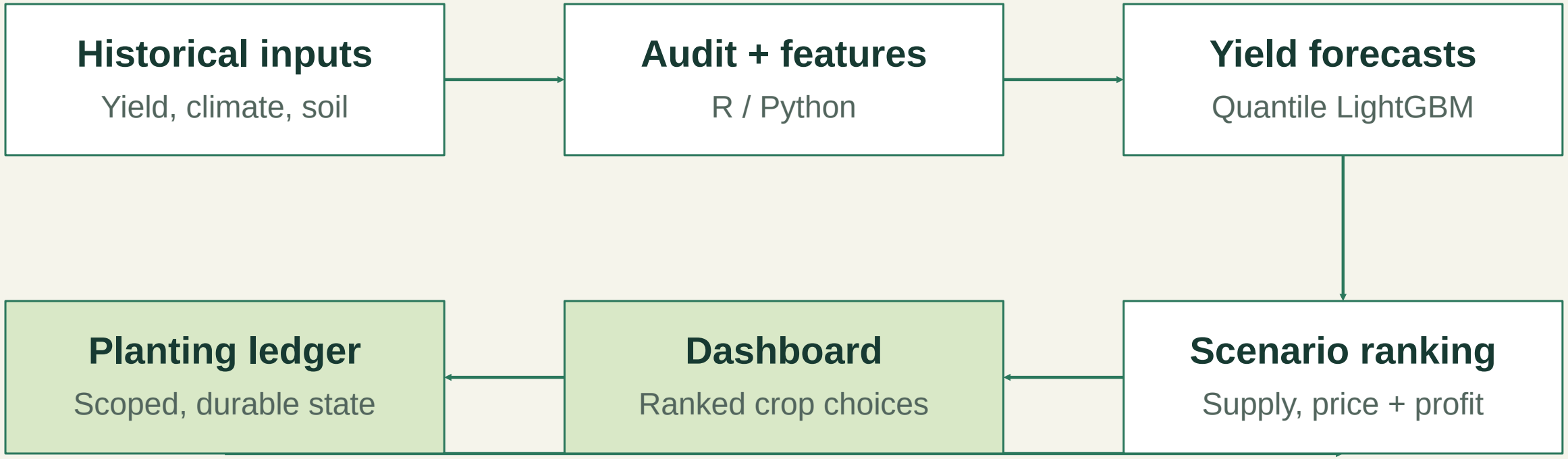
04 **Compare against simple baselines**

Measure held-out error, interval coverage and ranking quality.

Scope excludes nationwide validated forecasts, personalized farm yields and guaranteed income gains.

Proposed regional architecture

R / Python pipeline, FastAPI service and React dashboard



New commitments update supply and subsequent advice.

Price scenarios include an estimate of supply outside the application.

Dataset and data preparation

District–crop–season records with starting years 2000–2019

95,674
raw records

95,667
cleaned records

212
state–district pairs

45
crop types

Inputs

Historical yield, rainfall, irrigation and soil indices
State, district, crop and planting season

Target

Yield in tonnes per hectare

Madhya Pradesh **Maharashtra** **Punjab**
Tamil Nadu **Uttar Pradesh**

Direct source: project CSV. Economic prices and costs include modeled proxies.

Information available before sowing

Historical isolation is part of preprocessing, not just the train/test split

Key integrity

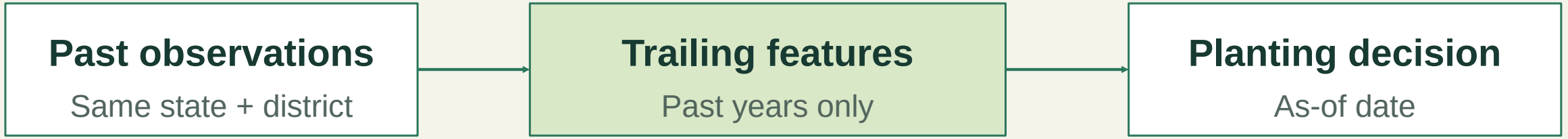
Remove duplicates and resolve colliding records.

Missing values

Use training-period imputation and preserve flags.

Physical consistency

Check units, irrigation denominators and yield ranges.

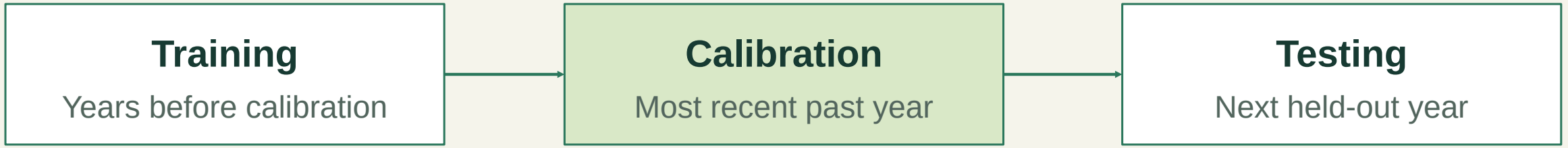


Group keys include state. Future records must not change earlier feature values.

Model development and validation

LightGBM quantile models handle nonlinear relationships in tabular data

Lower 10% model **Median 50% model** **Upper 90% model**



Learning rate	Leaves	Max depth	Boosting rounds
0.05	31	8	500

Baseline: trailing three-year yield mean
Walk-forward reference folds: 2015–2019

Yield and price become profit scenarios

One hectare: yield in tonnes, price in rupees per quintal

Scenario profit = 10 × yield × price − cultivation cost

The factor 10 converts tonnes to quintals.

Central scenario

Median yield and median price, with costs known before sowing

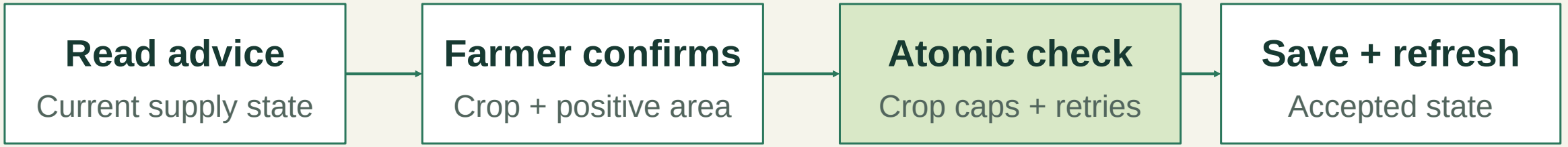
Downside scenario

A conservative yield–price combination with explicit assumptions

Scenario score = central profit − 0.5 × downside gap

Commitments change subsequent advice

A voluntary commitment updates regional supply state



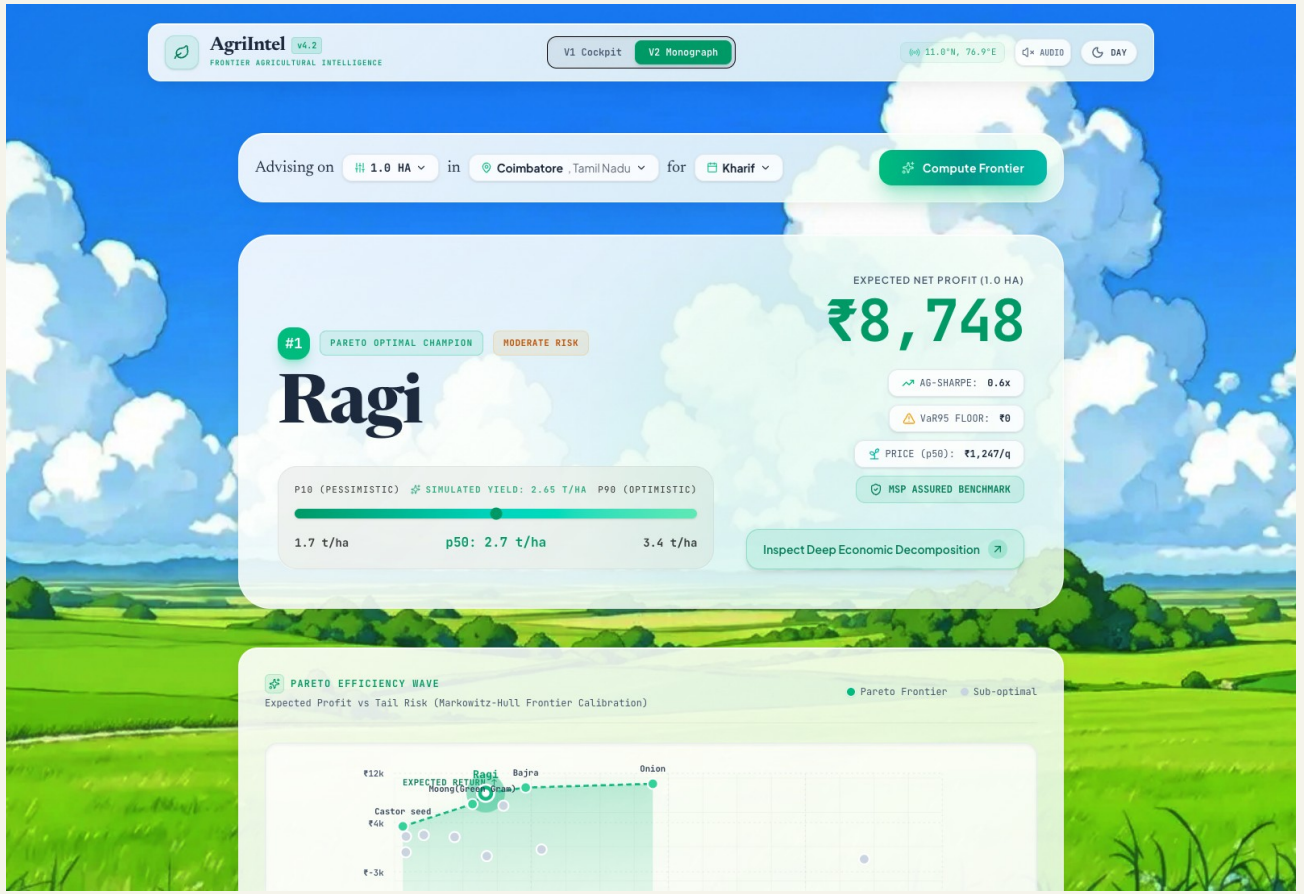
Illustrative onion capacity example

Capacity	Committed	Remaining	New request
700 ha	600 ha	100 ha	200 ha

Request exceeds remaining capacity. Offer a supported alternative crop.

Application and representative output

Existing interface captured from this project, with a saved 2019 scenario



Tamil Nadu
Coimbatore
Kharif, 1 hectare

Ragi

2.65 t/ha median yield

₹8,748 central profit

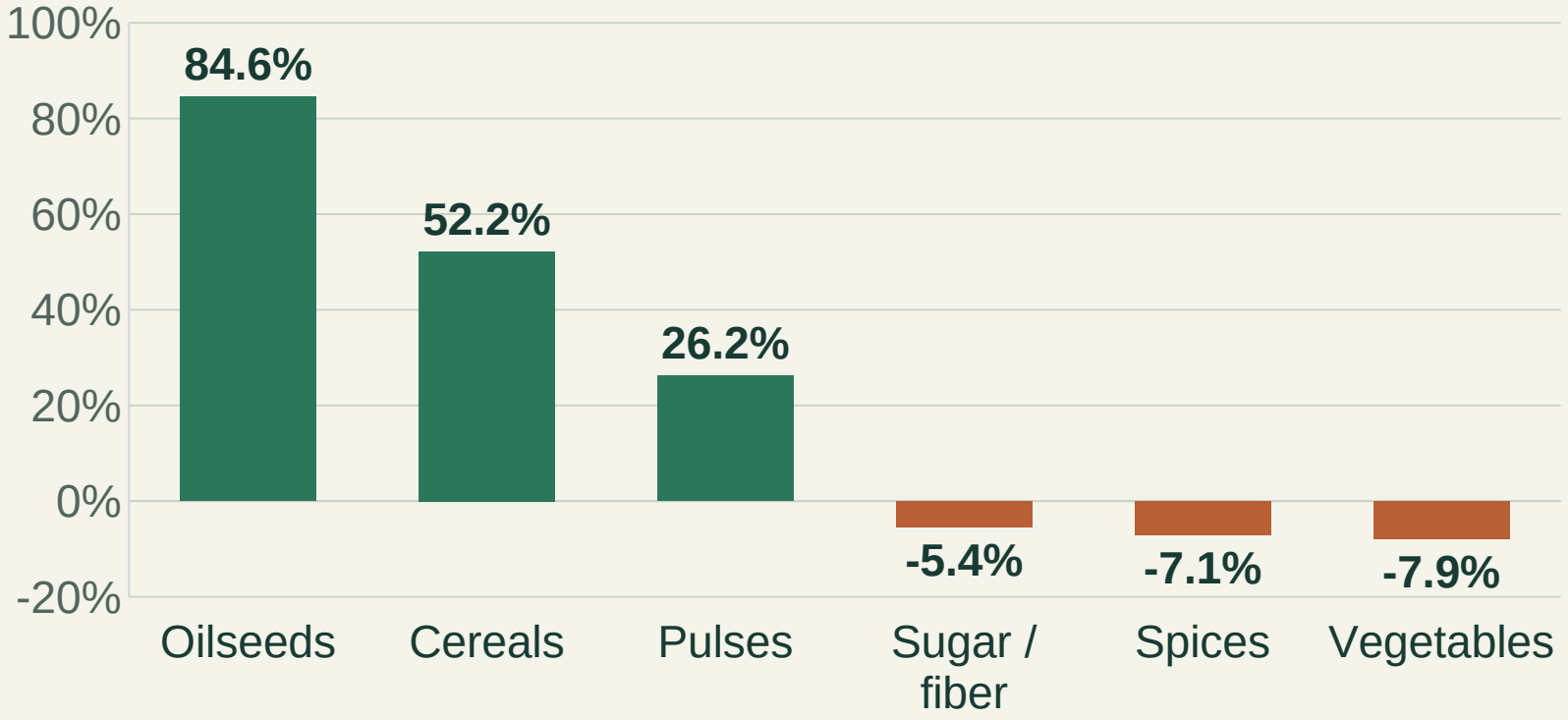
–₹4,741 pessimistic profit

Existing UI retains legacy labels. Values here are modeled scenarios, not assured income.

Historical yield results

Saved walk-forward backtest outputs for 2015–2019

RMSE reduction relative to baseline



Overall RMSE (t/ha)

Baseline 4.6048

Model 4.6189

78.09%

interval coverage

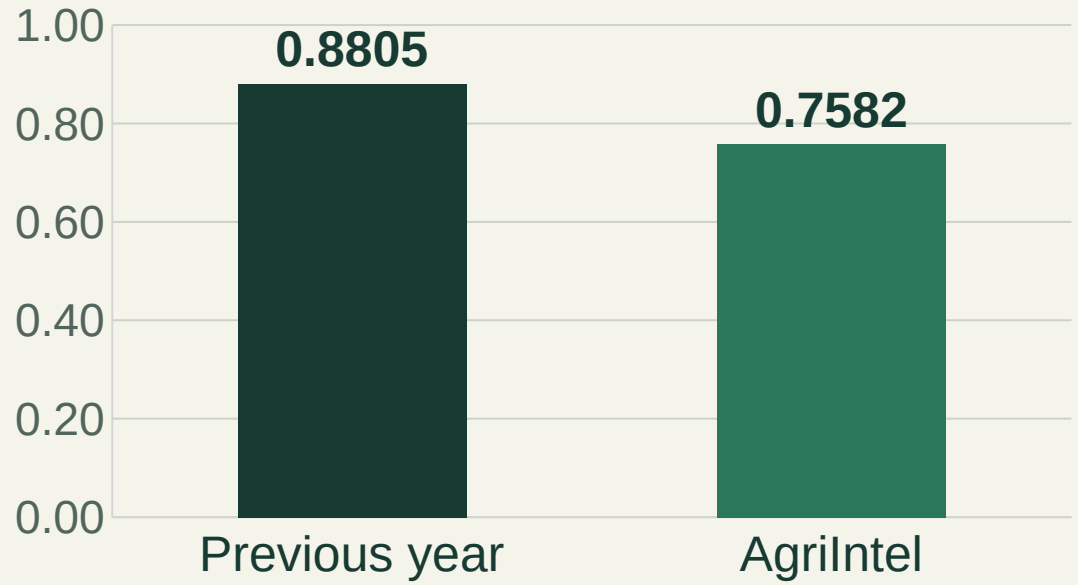
80% nominal target

Best saved overall point predictor: baseline. Prospective accuracy needs further evaluation.

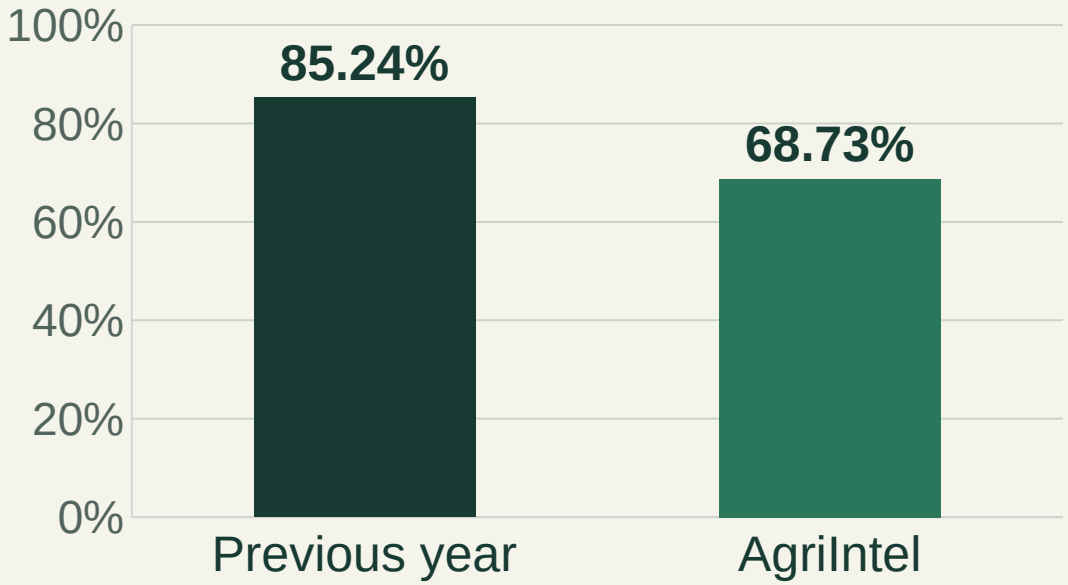
Historical recommendation results

Targets use simulated prices and cultivation-cost proxies

NDCG@3



Top choice in realized top 3



The previous-year baseline leads on both saved ranking metrics.

Historical reference only. These results do not demonstrate improved farmer income.

Challenges and safeguards

Reliability requirements for the regional workflow

Challenge	Safeguard
Future information	Past-only features and train-period statistics
Concurrent commitments	Atomic caps, positive area and safe retries
Costs and units	One score with consistent cost and units
Stale or missing data	Visible data date and unavailable-data message

Engineering gate: no violations in historical-isolation, capacity, retry/recovery and unit-reconciliation checks.

Conclusion and future scope

Regional advisory design

A regional workflow with safeguarded planting commitments and transparent yield and financial scenarios

Key finding

Historical performance is mixed. Prospective accuracy and actual farmer income effects require new evaluation.

National expansion

Audit crop–region coverage across 28 states and 8 UTs.
Publish predictions only where data and validation support them.

References

Project evidence, methods and data-source documentation

Project data and saved results

agriIntel_merged_final_v.csv
M0 audit, M3 yield and M5 ranking manifests

Project implementation

config/pipeline.yaml
modules/ and backend/main.py

Official agricultural statistics

Directorate of Economics and Statistics
data.desagri.gov.in

Climate and inflation sources

IMD gridded rainfall / imdpune.gov.in
World Bank CPI / indicator FP.CPI.TOTL

Model and application libraries

LightGBM / lightgbm.readthedocs.io
FastAPI / fastapi.tiangolo.com React / react.dev

Prediction feedback research

Perdomo et al. (2020), Performative Prediction
ICML, Proceedings of Machine Learning Research 119

Full source paths, URLs and evidence qualifications appear in the speaker notes.